UI/UX DESIGN & PRODUCT DESIGN · TASK 1

User Research & Problem Discovery

UI/UX fundamentals, Design Thinking, user research, competitor analysis, personas, empathy map

12	4	2	6
survey questions	alternatives compared	personas	measurable goals

INTERN
Eswar Mahalingam

PROGRAMME
Sumerix Global · UI/UX Design · 36 weeks

BATCH START
15 October 2026

PRODUCT
ShiftLoop — hub shift-handover app

1 · Executive summary

1
validated problem space: hub shift handover

12 + 9
survey questions + interview prompts

4
alternatives analysed (3 products + status quo)

2
personas + 1 empathy map

ShiftLoop is a mobile app for logistics gateway hubs that turns the end-of-shift handover into a guided, offline-first, 6-section checklist with photo and voice evidence and an explicit acknowledgement by the incoming lead. This report covers Task 1 of the internship: UI/UX fundamentals, Design Thinking, user research design, competitor analysis, personas, empathy map, problem definition, goals and the design workspace.

Research integrity note. The brief requires ≥ 5 interviews or survey responses. The questionnaire, interview guide and analysis sheet are ready (research/). The charts in section 5 use an **8-row pilot dataset that is DECLARED SYNTHETIC** — created only to build and test the analysis — and must be replaced with real responses from hub colleagues before any conclusion is quoted. Personas are therefore labelled *proto-personas*.

2 · Part A — UI vs UX with real-world apps

User Interface (UI) is what you see and touch: layout, colour, type, icons, buttons, motion. In **WhatsApp** the UI is the green header, chat bubbles, tick marks and the floating new-chat button.

User Experience (UX) is how well the whole journey works for a goal: finding a chat, sending a voice note with one hand, knowing a message was read (blue ticks), working on a weak 2G signal.

App	UI example	UX example	Lesson for ShiftLoop
WhatsApp	Bubbles, ticks, voice-note mic	Read receipts remove 'did they see it?' anxiety	Show acknowledgement state on every critical item
Amazon	Product cards, sticky Buy Now	One-click reorder; order tracking reduces calls	Pre-fill from scans so leads confirm instead of type
Spotify	Dark theme, album art	Offline downloads for the commute	Offline-first: never lose a handover at the dock

Product Design adds business viability and strategy (what to build and why); **Service Design** covers the whole service across channels — e.g. the huddle, the notice board, the IT desk and the app together.

3 · Part B — Design Thinking applied to the handover problem

Stage	What we do for ShiftLoop	Output

Empathize	Observe 3 live handovers (one per shift), interview leads, associates and the hub manager	Field notes, quotes, survey responses
Define	Synthesise into personas, empathy map and a problem statement with HMW questions	Problem statement (section 8)
Ideate	Crazy-8s on the 5-minute handover; card sort to name sections	Concepts, IA (Task 2)
Prototype	Lo-fi wireframes → hi-fi Figma screens → clickable prototype	Tasks 2–4
Test	Timed handover with 5 leads; acknowledge-flow test with incoming leads	Usability findings, iteration

The **Double Diamond** frames the same work: Discover/Define (Task 1) diverge then converge on the problem; Develop/Deliver (Tasks 2–6) diverge on solutions and converge on the shipped design.

4 · Part C — User research plan

Secondary research

- India's logistics cost was estimated at **7.8–8.9% of GDP in 2021-22** (NCAER, Dec 2023) — small efficiency gains at hubs matter. source 1
- The UK HSE lists **shift handover** as a key human-factors topic: miscommunication at handover has contributed to serious incidents in process industries. source 2
- Frontline apps (Connecteam, Deputy, Microsoft Shifts) focus on scheduling, time and chat — not on operational handover content. sources 3–5

Primary research design

- **Survey** — 12 questions (Google Forms), target 15 responses across 3 shifts
- **Interviews** — 5 × 30 min (2 night leads, 1 morning lead, 1 associate, 1 hub manager)
- **Observation** — 3 live handovers with a timing & tools checklist
- **Ethics** — anonymous, opt-in, no photos of people or customer data

Survey questionnaire (full version in research/survey_questionnaire.md)

#	Question	Type
Q1	What is your role at the hub?	Single choice
Q2	Which shift do you usually work?	Single choice
Q3	How is information passed to the next shift today?	Multiple choice
Q4	How many minutes does a typical handover take?	Number
Q5	In the last 2 weeks, how often did you start a shift without knowing something important?	Scale 0–5+
Q6	What went wrong the last time a handover failed? Describe what happened.	Open text
Q7	Which problems do you face during handover?	Multiple choice
Q8	What do you do when information is missing?	Open text
Q9	Rate: 'I trust the information I receive at shift start.'	Likert 1–5

Q10	Which device do you use on the floor?	Single choice
Q11	Which language do you prefer for work apps?	Single choice
Q12	If one thing about handovers could change tomorrow, what would it be?	Open text

Interview guide

1. Warm-up (2 min): role, hub, shift pattern, years in logistics.
2. Walk me through the last handover you GAVE — start to finish. (observe tools, time, people)
3. Walk me through the last handover you RECEIVED. What did you check first?
4. Tell me about a time something important was not passed on. What happened next? Who fixed it?
5. Where does shift information live today? (show me the WhatsApp group / register if allowed)
6. What do you do when the network is down at the dock?
7. Who is accountable when an exception is missed? How is that decided?
8. If you had a magic button for handover, what would it do?
9. Wrap-up: anything we didn't ask? Can we observe a live handover?

5 · Pilot analysis (DECLARED SYNTHETIC — replace with real responses)

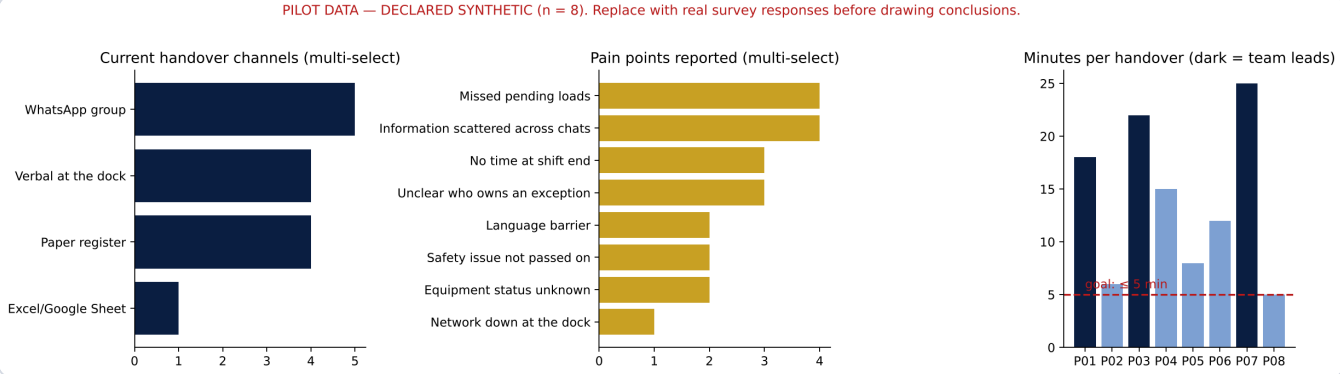


Figure 1 — The analysis pipeline built on 8 synthetic pilot rows (research/pilot_responses_DECLARED_SYNTHETIC.csv). Same script/columns accept the real Google-Forms export.

Hypotheses to validate with the real survey: H1 WhatsApp is the main channel for leads · H2 'information scattered' and 'missed pending loads' are the top pain points · H3 lead handovers exceed 10 minutes · H4 trust in received information is ≤ 3/5.

6 · Part D — Competitor analysis

Competitor analysis — shift handover at a logistics hub

Heuristic review against the brief's criteria. Ratings are the designer's assessment of public product pages and store listings (Nov 2026) — validate hands-on during interviews.

Criterion	Connecteam	Deputy	Microsoft Teams — Shifts	Status quo: WhatsApp group + paper register
Primary job	All-in-one frontline app: scheduling, time clock, checklists/forms, chat	Scheduling, timesheets and labour cost	Rosters and shift swaps inside Teams	Free-form chat + handwritten log
Navigation	4/5 Tab bar + module hub; many modules to discover	4/5 Clear schedule-first navigation	3/5 Lives inside Teams; several taps to reach Shifts	2/5 Chronological chat only; no structure
Layout & visual hierarchy	4/5 Card-based, configurable	4/5 Clean calendar/timeline views	3/5 Dense Office styling	1/5 Wall of messages; important items scroll away
Search	3/5 Search inside modules	3/5 People/shift search	4/5 Teams search across chats/files	2/5 Keyword search in chat; register not searchable
'Checkout' = handover sign-off	3/5 Possible via custom checklist/form; not hub-specific	2/5 No dedicated handover template found on public pages	2/5 Notes per shift; no structured acknowledgement	1/5 No sign-off, no owner, no timestamp
Offline use at the dock	3/5 Varies by module	3/5 Some offline features	2/5 Needs connectivity for most actions	4/5 Paper always works; chat queues
Accessibility & language	4/5 Multiple languages; readable type	3/5 Good contrast; limited Indian languages	4/5 Office accessibility tooling	3/5 Familiar UI; voice notes help low-literacy users
Overall UX for a hub handover	3/5 Powerful but generic — needs set-up per site	2/5 Strong for rostering, weak for operational context	2/5 Fits office teams better than floor teams	2/5 Fast to start, fails on accountability

Strengths & weaknesses

Product	Strengths	Weaknesses / opportunity for ShiftLoop
Connecteam	Checklists, forms, chat and scheduling in one app; frontline focus	Generic forms → each hub must design its own handover; can feel heavy for associates
Deputy	Mature scheduling, compliance and timesheet flows	Handover context (pending loads, exceptions, dock status) is outside its core model
MS Teams — Shifts	Already in many enterprises; strong identity & search	Office-centric UI; floor staff rarely live in Teams; weak offline story
WhatsApp + register	Zero training, works on any phone, voice notes	No structure, no ownership, no audit trail; safety items get buried

Design opportunity: none of the options combines a hub-specific checklist, explicit owner + acknowledgement, offline-first capture and voice/photo evidence in one 5-minute flow.

Figure 2 — Heuristic comparison against the brief's criteria (navigation, layout, search, 'checkout' = sign-off, accessibility, visual hierarchy, UX). Ratings are the designer's assessment of public pages (Nov 2026) and are re-checked hands-on during research.

7 · Parts E & F — Personas and empathy map

DK

Divya Krishnan

29 · Night-shift Team Lead · Gateway hub, Coimbatore

"By 5:45 I'm fighting three fires. The handover is the last thing I want to type into WhatsApp."

Digital confidence

Time pressure

Need for accountability

Proto-persona: built from secondary research and the designer's floor experience; validate with ≥ 5 interviews.

GOALS

- Close the shift with zero surprises for the morning lead
- Prove her team hit SLA despite late linehauls
- Spend < 5 minutes on handover

PAIN POINTS

- Pending loads and exceptions scattered across 4 WhatsApp groups
- Network drops inside the dock
- No record of who acknowledged a safety flag
- Types everything twice (register + chat)

TECHNOLOGY USAGE

Mid-range Android, WhatsApp power user, basic Excel, uses voice notes in Tamil and English

MOTIVATION

Recognition as a reliable lead; promotion to hub manager

DIGITAL SKILLS

Medium — confident with apps she uses daily, avoids long forms

RI

Rajesh Iyer

44 · Hub Manager · accountable for 3 shifts and SLA

"When a bag is missed at 8 a.m., I need to know in one minute what the night shift knew."

Digital confidence

Time pressure

Need for accountability

Proto-persona: built from secondary research and the designer's floor experience; validate with ≥ 5 interviews.

GOALS

- Audit trail of every handover and exception
- Spot recurring issues (docks, equipment) across weeks
- Fewer escalations from customers and linehaul partners

PAIN POINTS

- Reconstructs incidents from chat screenshots
- Cannot compare shifts objectively
- Safety issues reported late

TECHNOLOGY USAGE

Laptop + Android, Excel/Power BI dashboards, email-heavy

MOTIVATION

Hub ranking, zero safety incidents

DIGITAL SKILLS

High — comfortable with dashboards and reports

DK **Divya · Night-shift Team Lead** at shift handover (05:30–06:15)

THINKS

- "If I miss this, the morning shift will blame my team."
- "Typing a long message now will delay dispatch."
- "Did anyone even read my safety note?"

FEELS

- Pressured in the last 30 minutes of the shift
- Anxious about un-owned exceptions
- Proud when the next shift starts clean

SAYS

- "Check the group, I posted it."
- "Bag 4471 is pending — tell Arun."
- "Forklift on D7 is down again."

DOES

- Sends 10–20 WhatsApp messages and voice notes
- Writes a short note in the paper register
- Walks the docks for a last check, phone in hand

PAIN POINTS

- Information scattered across chats and paper
- No acknowledgement or owner
- Network drops inside the dock
- Same data typed twice

GAINS

- Handover done in < 5 minutes
- Proof that every critical item was acknowledged
- Fewer calls after going home
- Recognition for clean shifts

Figure 3 — Empathy map for Divya at the moment of handover (05:30–06:15).

Top 5 pain points ShiftLoop must address

1. Information scattered across chats, paper and memory
2. No owner or acknowledgement for critical items
3. Network drops inside docks

4. Re-typing the same facts twice
5. Safety flags reported late or not at all

8 · Problem statement & project goals

Problem statement. Night- and evening-shift team leads at gateway hubs need a fast, structured way to hand over pending loads, exceptions, safety and equipment status to the incoming lead, because today that information is scattered across WhatsApp groups, paper registers and verbal briefings — so items are missed, nobody can prove who acknowledged a safety risk, and the next shift starts by firefighting instead of planning.

How might we…

- How might we let a tired team lead capture a complete handover in under 5 minutes, even without network?
- How might we make every critical item have an owner and a timestamped acknowledgement?
- How might we give hub managers an audit trail without asking leads to type twice?

Type	Goal	Success metric
User goal	Complete a handover in ≤ 5 min (median)	Timed in usability test & app analytics
User goal	Incoming lead knows every open item before the first truck	≥ 95% critical items acknowledged within 15 min
Business goal	Fewer missed pending loads / SLA breaches caused by handover gaps	−50% handover-related misses in 8-week pilot
Business goal	Auditable safety communication	100% of safety flags carry owner + acknowledgement
Product goal	Adoption by floor leads	≥ 80% of shifts in the pilot hub use ShiftLoop by week 4
Experience goal	Trust in received information	Trust score (Q9) from ≤ 2.5 → ≥ 4.0 on a 5-point scale

9 · Part G — Professional design workspace

```
ShiftLoop/                        (Figma team
project + this repo)
├── 00_Admin/          brief, timeline, daily log
├── 01_Research/       survey, interview notes,
synthesis
├── 02_IA_Flows/       sitemap, flows, card sort
(Task 2)
├── 03_Wireframes/     lo-fi frames
├── 04_UI/             design system + hi-fi (Task
3)
├── 05_Prototype/      interactive flows (Task 4)
└── 99_Archive/        superseded versions
```

Naming & versioning

- Files: `SL_T1_Research_Personas_v01`
- Frames: `SL/Mobile/08_Handover/Default`
- Components: `Button/Primary/Default`, `Chip/Status/Error`
- Version history: named Figma versions at each task submission (*T1-submitted*, *T2-submitted* …) + Git tags in this repo

- Asset library: icons (24 px), colour & text styles published from `04_UI/Design System`

Figma link: the project files in this repo (PNG/SVG screens) are ready to import (File ▸ Import). Add the Figma share link in README.md once the team file is created.

10 · Recommended product direction & next steps

Direction: a focused, offline-first handover checklist for team leads with acknowledgement by the incoming lead — not another all-in-one workforce app.

Next (Task 2): information architecture, user flows for register / login / handover / acknowledge / logout, journey map, card sort and 10+ low-fidelity wireframes.

Sources

1. NCAER (2023) Logistics Cost in India — https://www.ncaer.org/wp-content/uploads/2023/12/NCAER_Report_LogisticsCost2023.pdf
2. UK HSE — Human factors: shift handover — https://www.hse.gov.uk/humanfactors/topics/shift-handover.htm
3. Connecteam — shift scheduling app — https://connecteam.com/employee-scheduling-app/shift-scheduling/
4. Deputy — https://www.deputy.com/
5. Microsoft Support — Get started in Shifts — https://support.microsoft.com/en-us/office/get-started-in-shifts-5f3e30d8-1821-4904-be26-c3cd25a497d6